KRISHNA KOUSHIK MADDUKURI

SENIOR SOFTWARE ENGINEER

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PROFESSIONAL SUMMARY

Senior Software Engineer with 5+ years of experience designing, building, and operating large-scale distributed systems and cloud-native microservices in the financial services domain. Expert in Java, Spring Boot, Apache Kafka, and AWS, with a proven record of delivering secure, compliant, high-throughput data platforms at Capital One. Reduced batch processing time by 43%, cut deployment time by 70%, and lowered compute costs through container optimization. AWS Certified Developer with deep experience in CI/CD automation, event-driven architecture, REST API design, and production observability.

CORE TECHNICAL SKILLS

Languages	Java, Python, TypeScript, JavaScript, SQL, C++
Backend & Frameworks	Spring Boot, Spring MVC, Spring Batch, Spring Kafka, Hibernate, JPA, Microservices, REST APIs, GraphQL, Node.js, OAuth2, Liquibase
Cloud & AWS	AWS (ECS, Fargate, EC2, S3, SNS, Lambda, RDS, DynamoDB, Step Functions, CloudFormation, CloudWatch, Glue, EMR), Docker, Kubernetes, Serverless Architecture
Data & Messaging	Apache Kafka, PostgreSQL, DynamoDB, Event-Driven Architecture, Real-Time Data Pipelines, ETL
DevOps & CI/CD	Jenkins, Git, Bitbucket, Maven, SonarQube, Blue-Green Deployments, Infrastructure as Code (IaC)
Testing & Quality	JUnit, Mockito, Cucumber, Selenium, WireMock, Postman, Test-Driven Development (TDD)
Observability	Splunk, New Relic APM, Prometheus, ELK Stack, Spring Actuator, Log4J
Frontend	React.js, Redux, Angular, HTML5, CSS3, Bootstrap, jQuery

PROFESSIONAL EXPERIENCE

Senior Software Engineer — Capital One

Sep 2024 – Present · McLean, VA

- Designed and built scalable, secure Java/Spring Boot microservices for Capital One's Enterprise Credit Bureau Reporting Service, generating compliant Metro2 files for U.S., Canada, Small Business, and Commercial segments across Equifax, Experian, TransUnion, and LexisNexis.
- Engineered high-throughput, event-driven pipelines with Apache Kafka, Spring Kafka, and PostgreSQL/RDS, achieving fault-tolerant, exactly-once processing for sensitive financial workflows at enterprise scale.
- Improved batch performance by 43% (7 hours to 4) by replacing cursor-based readers with a paginated reader in Spring Batch for high-volume Metro2 file generation.
- Deployed containerized services on AWS Fargate via CloudFormation, integrating SNS, S3, RDS, and Step Functions; reduced container size by 60%, lowering compute costs and improving runtime efficiency.
- Implemented enterprise CI/CD pipelines with Jenkins to automate builds, enforce SonarQube quality gates, and manage blue-green deployments with zero-downtime releases.
- Strengthened security and compliance by designing OAuth2-protected REST APIs and building automated test suites with JUnit and Cucumber, reducing production defects.
- Integrated Splunk and New Relic APM with actionable dashboards and alerting, accelerating incident response and root-cause analysis across distributed microservices.

Software Engineer — NGA Group Inc Sep 2023 – Sep 2024 · New Brunswick, NJ

- Designed a cloud-ready Spring Boot service unifying disparate data sources into a standardized reporting model, improving data accessibility and eliminating manual reconciliation across analytics workflows.
- Delivered secure, role-based REST APIs enabling downstream teams to consume curated datasets without direct database access, strengthening data governance.
- Built a high-velocity ingestion pipeline with Apache Kafka and DynamoDB, supporting near-real-time propagation of updates and scaling with increased upstream event volume.
- Containerized the application on AWS ECS and automated build/release pipelines through Jenkins, reducing deployment time by 70% and improving release reliability.
- Improved operational visibility by integrating Spring Actuator, Prometheus, and ELK analytics, cutting issue-detection time by 50%.

Software Engineer — Infosys Limited Aug 2020 – Aug 2021 · Hyderabad, India

- Developed modular, reusable React components with Redux state management, delivering consistent UI behavior across 4 large-scale applications.
- Integrated RESTful APIs with React and Redux for asynchronous data fetching, improving client-server communication and content rendering.
- Built frontend features — forms, routing, events, custom components — integrated with Spring MVC backend services for reliable API orchestration.
- Modernized legacy codebases and applied engineering best practices, reducing bug rates by 25% and improving maintainability and performance.
- Implemented CI/CD pipelines with Bitbucket and automated multi-region AWS deployments, increasing deployment frequency by 3×.

Software Engineer Intern — Infosys Limited Jan 2020 – Aug 2020 · Mysore, India

- Completed 15+ projects and certification-aligned exercises across back-end and front-end development within the MERN ecosystem.
- Diagnosed and resolved backend build failures by analyzing Jenkins logs and debugging deployment scripts in Bash and Groovy, improving CI/CD pipeline stability by 40%.
- Coordinated sprint planning for a four-developer team using Agile methodology and JIRA, ensuring on-time delivery of project milestones.

EDUCATION

Doctorate in Business Administration (DBA), Technology Management — Westcliff University Jul 2026 – Aug 2029 (expected)

Master of Science in Computer Science — George Mason University, Fairfax, VA 2023

Bachelor of Technology in Computer Science — Amrita Vishwa Vidyapeetham, Coimbatore, India 2020

CERTIFICATIONS

- AWS Certified Developer - Associate
- AWS Certified Cloud Practitioner